SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : Case Study (Medium)
Assessment Tool Weightage: 30%

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	**5**
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	**5**
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	**5**
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	**5**
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	**5**
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	**5**

7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	**5**
8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	**5**
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	**5**
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	**5**

Code of Conduct:

*I **KARNAM HARSHITH** - **192511384** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
RUBRICS FOR CALCULATION:

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1) Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.

Given Relation

ORDER(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)

Assume:

- OrderID → Order number
- CustID → Customer ID
- CustName → Customer name
- ProdID → Product ID
- ProdName → Product name
- Qty → Quantity ordered
- Price → Product price

Functional Dependencies

- CustID → CustName
- ProdID → ProdName, Price
- (OrderID, ProdID) → Qty

The candidate key is:

(OrderID,ProdID)

because one order can contain multiple products.

Identify Dependencies
There is a partial dependency:

OrderID, ProdID → Qty
ProdID → ProdName, Price

Since ProdName and Price depend only on ProdID, they depend on part of the composite key.

There are also dependencies involving customer information:

CustID → CustName
So we decompose the relation to remove redundancy and achieve 3NF.

3NF Decomposition

1. CUSTOMER
CUSTOMER(CustID, CustName)

2. PRODUCT
PRODUCT(ProdID, ProdName, Price)

3. ORDER
ORDER(OrderID, CustID)

4. ORDER_ITEM
ORDER_ITEM(OrderID, ProdID, Qty)

CREATE TABLE AND INSERT DATA

```
Database changed
mysql> CREATE TABLE Customer (
    ->     CustID INT PRIMARY KEY,
    ->     CustName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Customer
    -> VALUES
    -> (1, 'Rahul'),
    -> (2, 'Priya'),
    -> (3, 'Arun');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Product (
    ->     ProdID INT PRIMARY KEY,
    ->     ProdName VARCHAR(50),
    ->     Price DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Product
    -> VALUES
    -> (101, 'Laptop', 65000),
    -> (102, 'Mouse', 800),
    -> (103, 'Keyboard', 1500);
Query OK, 3 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE Orders (
    ->     OrderID INT PRIMARY KEY,
    ->     CustID INT,
    ->     FOREIGN KEY (CustID) REFERENCES Customer(CustID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Orders
    -> VALUES
    -> (1001, 1),
    -> (1002, 2),
    -> (1003, 3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Order_Item (
    ->     OrderID INT,
    ->     ProdID INT,
    ->     Qty INT,
    ->     PRIMARY KEY (OrderID, ProdID),
    ->     FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
    ->     FOREIGN KEY (ProdID) REFERENCES Product(ProdID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Order_Item
    -> VALUES
    -> (1001, 101, 1),
    -> (1001, 102, 2),
    -> (1002, 103, 1),
    -> (1003, 102, 3);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Customer;
+--------+----------+
| CustID | CustName |
+--------+----------+
|      1 | Rahul    |
|      2 | Priya    |
|      3 | Arun     |
+--------+----------+
3 rows in set (0.01 sec)

mysql> SELECT * FROM Product;
+--------+----------+----------+
| ProdID | ProdName | Price    |
+--------+----------+----------+
|    101 | Laptop   | 65000.00 |
|    102 | Mouse    |   800.00 |
|    103 | Keyboard |  1500.00 |
+--------+----------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Orders;
+---------+--------+
| OrderID | CustID |
+---------+--------+
|    1001 |      1 |
|    1002 |      2 |
|    1003 |      3 |
+---------+--------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Order_Item;
+---------+--------+------+
| OrderID | ProdID | Qty  |
+---------+--------+------+
|    1001 |    101 |    1 |
|    1001 |    102 |    2 |
|    1002 |    103 |    1 |
|    1003 |    102 |    3 |
+---------+--------+------+
4 rows in set (0.00 sec)

mysql> SELECT
    ->     O.OrderID,
    ->     C.CustName,
    ->     P.ProdName,
    ->     OI.Qty,
    ->     P.Price
    -> FROM Orders O
    -> JOIN Customer C
    ->     ON O.CustID = C.CustID
    -> JOIN Order_Item OI
    ->     ON O.OrderID = OI.OrderID
    -> JOIN Product P
    ->     ON OI.ProdID = P.ProdID;
+---------+----------+----------+------+----------+
| OrderID | CustName | ProdName | Qty  | Price    |
+---------+----------+----------+------+----------+
|    1001 | Rahul    | Laptop   |    1 | 65000.00 |
|    1001 | Rahul    | Mouse    |    2 |   800.00 |
|    1002 | Priya    | Keyboard |    1 |  1500.00 |
|    1003 | Arun     | Mouse    |    3 |   800.00 |
+---------+----------+----------+------+----------+
4 rows in set (0.01 sec)
```

Justification
The original relation contains redundant customer and product information because the same customer and product details can appear in many orders. Decomposing it into Customer, Product, Orders, and Order_Item removes partial and transitive dependencies while preserving the relationships. Hence, the resulting schema satisfies 3NF.

2) A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.

Given Relation

PATIENT(PID, PName, DocID, DocName, Specialization, WardNo, WardName)
Assume the following functional dependencies:

PID → PName, DocID, WardNo

DocID → DocName, Specialization

WardNo → WardName

Therefore, there are transitive dependencies:
PID → DocID → DocName, Specialization
And

PID → WardNo → WardName
Identify Anomalies
1. Update Anomaly
If a doctor's name changes, it must be updated in multiple patient records.

For example, if DocID = D01 appears for 20 patients, the doctor's name is repeated 20 times.

2. Insertion Anomaly
A new doctor cannot easily be added unless a patient is assigned to that doctor.

3. Deletion Anomaly
If the last patient assigned to a doctor is deleted, the doctor's information may also be lost.

BCNF Analysis
A relation is in BCNF when every determinant in a non-trivial functional dependency is a superkey.

In the original relation:

DocID → DocName, Specialization

But DocID is not a superkey of the complete Patient relation.

Similarly:

WardNo → WardName

but WardNo is not a superkey of the complete relation.

Therefore:
Patient is not in BCNF

BCNF DECOMPOSITION
Separate the determinants into their own relations.

1. PATIENT

PATIENT(PID, PName, DocID, WardNo)

2. DOCTOR

DOCTOR(DocID, DocName, Specialization)

3. WARD

WARD(WardNo, WardName)

```
mysql> CREATE TABLE Doctor (
    ->     DocID VARCHAR(10) PRIMARY KEY,
    ->     DocName VARCHAR(50),
    ->     Specialization VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Doctor
    -> VALUES
    -> ('D01', 'Kumar', 'Cardiology'),
    -> ('D02', 'Priya', 'Neurology'),
    -> ('D03', 'Ravi', 'Orthopedics');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Ward (
    ->     WardNo INT PRIMARY KEY,
    ->     WardName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Ward
    -> VALUES
    -> (101, 'General Ward'),
    -> (102, 'ICU'),
    -> (103, 'Emergency');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Patient (
    ->     PID INT PRIMARY KEY,
    ->     PName VARCHAR(50),
    ->     DocID VARCHAR(10),
    ->     WardNo INT,
    ->     FOREIGN KEY (DocID) REFERENCES Doctor(DocID),
    ->     FOREIGN KEY (WardNo) REFERENCES Ward(WardNo)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Patient
    -> VALUES
    -> (1, 'Rahul', 'D01', 101),
    -> (2, 'Priya', 'D02', 102),
    -> (3, 'Arun', 'D01', 101),
    -> (4, 'Sneha', 'D03', 103);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Patient;
+-----+-------+-------+--------+
| PID | PName | DocID | WardNo |
+-----+-------+-------+--------+
|   1 | Rahul | D01   |    101 |
|   2 | Priya | D02   |    102 |
|   3 | Arun  | D01   |    101 |
|   4 | Sneha | D03   |    103 |
+-----+-------+-------+--------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Doctor;
+-------+---------+----------------+
| DocID | DocName | Specialization |
+-------+---------+----------------+
| D01   | Kumar   | Cardiology     |
| D02   | Priya   | Neurology      |
| D03   | Ravi    | Orthopedics    |
+-------+---------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Ward;
+--------+-------------+
| WardNo | WardName    |
+--------+-------------+
|    101 | General Ward |
|    102 | ICU          |
|    103 | Emergency    |
+--------+-------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     P.PID,
    ->     P.PName,
    ->     D.DocID,
    ->     D.DocName,
    ->     D.Specialization,
    ->     W.WardNo,
    ->     W.WardName
    -> FROM Patient P
    -> JOIN Doctor D
    ->     ON P.DocID = D.DocID
    -> JOIN Ward W
    ->     ON P.WardNo = W.WardNo;
+-----+-------+-------+---------+----------------+--------+-------------+
| PID | PName | DocID | DocName | Specialization | WardNo | WardName    |
+-----+-------+-------+---------+----------------+--------+-------------+
|   1 | Rahul | D01   | Kumar   | Cardiology     |    101 | General Ward |
|   3 | Arun  | D01   | Kumar   | Cardiology     |    101 | General Ward |
|   2 | Priya | D02   | Priya   | Neurology      |    102 | ICU          |
|   4 | Sneha | D03   | Ravi    | Orthopedics    |    103 | Emergency    |
+-----+-------+-------+---------+----------------+--------+-------------+
4 rows in set (0.00 sec)
```

Justification

The original relation has redundancy because doctor and ward information is repeated for multiple patients. DocID and WardNo determine non-key attributes but are not superkeys of the original relation, causing BCNF violations. Separating Doctor and Ward into independent relations removes these dependencies and produces a BCNF design.

3) Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.

Given Relation

ENROLLMENT(SID, SName, CourseID, CourseName, Faculty, Grade)

Functional Dependencies

SID → SName

CourseID → CourseName, Faculty

(SID, CourseID) → Grade

Candidate Key

- A student can enroll in multiple courses, so SID alone is not enough.
- A course can have multiple students, so CourseID alone is not enough.

Therefore:

(SID,CourseID)

is the candidate key.

Partial Dependencies

We have:

SID → SName
CourseID → CourseName, Faculty

Both depend on parts of the composite key, so the relation is not in 2NF.

2NF DECOMPOSITION

Separate the attributes that depend only on part of the key.

STUDENT

STUDENT(SID, SName)

COURSE

COURSE(CourseID, CourseName, Faculty)

ENROLLMENT

ENROLLMENT(SID, CourseID, Grade)

CREATE TABLE AND INSERT DATA

```
Database changed
mysql> CREATE TABLE Student (
    ->     SID INT PRIMARY KEY,
    ->     SName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Student
    -> VALUES
    -> (101, 'Rahul'),
    -> (102, 'Priya'),
    -> (103, 'Arun');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Course (
    ->     CourseID VARCHAR(10) PRIMARY KEY,
    ->     CourseName VARCHAR(50),
    ->     Faculty VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Course
    -> VALUES
    -> ('C01', 'DBMS', 'Dr. Kumar'),
    -> ('C02', 'Python', 'Dr. Priya'),
    -> ('C03', 'Java', 'Dr. Ravi');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Enrollment (
    ->     SID INT,
    ->     CourseID VARCHAR(10),
    ->     Grade VARCHAR(5),
    ->     PRIMARY KEY (SID, CourseID),
    ->     FOREIGN KEY (SID) REFERENCES Student(SID),
    ->     FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Enrollment
    -> VALUES
    -> (101, 'C01', 'A'),
    -> (101, 'C02', 'B+'),
    -> (102, 'C01', 'A+'),
    -> (102, 'C03', 'B'),
    -> (103, 'C02', 'A');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+-----+-------+
| SID | SName |
+-----+-------+
| 101 | Rahul |
| 102 | Priya |
| 103 | Arun  |
+-----+-------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Course;
+----------+------------+-----------+
| CourseID | CourseName | Faculty   |
+----------+------------+-----------+
| C01      | DBMS       | Dr. Kumar |
| C02      | Python     | Dr. Priya |
| C03      | Java       | Dr. Ravi  |
+----------+------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Enrollment;
+-----+----------+-------+
| SID | CourseID | Grade |
+-----+----------+-------+
| 101 | C01      | A     |
| 101 | C02      | B+    |
| 102 | C01      | A+    |
| 102 | C03      | B     |
| 103 | C02      | A     |
+-----+----------+-------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     S.SID,
    ->     S.SName,
    ->     C.CourseID,
    ->     C.CourseName,
    ->     C.Faculty,
    ->     E.Grade
    -> FROM Student S
    -> JOIN Enrollment E
    ->     ON S.SID = E.SID
    -> JOIN Course C
    ->     ON E.CourseID = C.CourseID;
+-----+-------+----------+------------+-----------+-------+
| SID | SName | CourseID | CourseName | Faculty   | Grade |
+-----+-------+----------+------------+-----------+-------+
| 101 | Rahul | C01      | DBMS       | Dr. Kumar | A     |
| 101 | Rahul | C02      | Python     | Dr. Priya | B+    |
| 102 | Priya | C01      | DBMS       | Dr. Kumar | A+    |
| 102 | Priya | C03      | Java       | Dr. Ravi  | B     |
| 103 | Arun  | C02      | Python     | Dr. Priya | A     |
+-----+-------+----------+------------+-----------+-------+
5 rows in set (0.00 sec)
```

Justification

The decomposition stores student and course details only once, reducing redundancy and avoiding update anomalies. The ENROLLMENT table retains the relationship between students and courses along with their grades, giving a proper 2NF structure.

4**) Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.**

Given Relation

BOOKING(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)

Assume these functional dependencies:

PassID → PassName

FlightNo → DeptCity, ArrCity, Date

(FlightNo, PassID) → Seat

The candidate key is:

(FlightNo,PassID)

because a passenger can book multiple flights and a flight can have multiple passengers.

Dependencies
We have:

PassID → PassName

and:

FlightNo → DeptCity, ArrCity, Date

Both depend on part of the composite key.

Therefore, the original relation is not even in 2NF.

Decompose it into:

PASSENGER

PASSENGER(PassID, PassName)

FLIGHT

FLIGHT(FlightNo, DeptCity, ArrCity, Date)

BOOKING

BOOKING(FlightNo, PassID, Seat)

Now each determinant is a key in its respective relation, so the resulting relations satisfy BCNF.

CREATE TABLE AND INSERT

```
mysql> CREATE TABLE Passenger (
    ->     PassID INT PRIMARY KEY,
    ->     PassName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Passenger
    -> VALUES
    -> (1, 'Rahul'),
    -> (2, 'Priya'),
    -> (3, 'Arun');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Flight (
    ->     FlightNo VARCHAR(10) PRIMARY KEY,
    ->     DeptCity VARCHAR(50),
    ->     ArrCity VARCHAR(50),
    ->     FlightDate DATE
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Flight
    -> VALUES
    -> ('F101', 'Chennai', 'Delhi', '2026-08-15'),
    -> ('F102', 'Mumbai', 'Bangalore', '2026-08-16'),
    -> ('F103', 'Hyderabad', 'Kolkata', '2026-08-17');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Booking (
    ->     FlightNo VARCHAR(10),
    ->     PassID INT,
    ->     Seat VARCHAR(10),
    ->     PRIMARY KEY (FlightNo, PassID),
    ->     FOREIGN KEY (FlightNo) REFERENCES Flight(FlightNo),
    ->     FOREIGN KEY (PassID) REFERENCES Passenger(PassID)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Booking
    -> VALUES
    -> ('F101', 1, '12A'),
    -> ('F101', 2, '12B'),
    -> ('F102', 1, '15A'),
    -> ('F103', 3, '10C');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Passenger;
+--------+----------+
| PassID | PassName |
+--------+----------+
|      1 | Rahul    |
|      2 | Priya    |
|      3 | Arun     |
+--------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Flight;
+----------+-----------+-----------+------------+
| FlightNo | DeptCity  | ArrCity   | FlightDate |
+----------+-----------+-----------+------------+
| F101     | Chennai   | Delhi     | 2026-08-15 |
| F102     | Mumbai    | Bangalore | 2026-08-16 |
| F103     | Hyderabad | Kolkata   | 2026-08-17 |
+----------+-----------+-----------+------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Booking;
+----------+--------+------+
| FlightNo | PassID | Seat |
+----------+--------+------+
| F101     |      1 | 12A  |
| F101     |      2 | 12B  |
| F102     |      1 | 15A  |
| F103     |      3 | 10C  |
+----------+--------+------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     B.FlightNo,
    ->     P.PassID,
    ->     P.PassName,
    ->     B.Seat,
    ->     F.DeptCity,
    ->     F.ArrCity,
    ->     F.FlightDate
    -> FROM Booking B
    -> JOIN Passenger P
    ->     ON B.PassID = P.PassID
    -> JOIN Flight F
    ->     ON B.FlightNo = F.FlightNo;
+----------+--------+----------+------+-----------+-----------+------------+
| FlightNo | PassID | PassName | Seat | DeptCity  | ArrCity   | FlightDate |
+----------+--------+----------+------+-----------+-----------+------------+
| F101     |      1 | Rahul    | 12A  | Chennai   | Delhi     | 2026-08-15 |
| F102     |      1 | Rahul    | 15A  | Mumbai    | Bangalore | 2026-08-16 |
| F101     |      2 | Priya    | 12B  | Chennai   | Delhi     | 2026-08-15 |
| F103     |      3 | Arun     | 10C  | Hyderabad | Kolkata   | 2026-08-17 |
+----------+--------+----------+------+-----------+-----------+------------+
4 rows in set (0.00 sec)
```

Justification
In each decomposed relation, the determinant is a candidate key: PassID determines passenger details and FlightNo determines flight details. The composite key (FlightNo, PassID) determines the seat in Booking, so there are no non-key determinants violating BCNF.

5) Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize it to 3NF.

Consider the HR relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName)

Functional Dependencies

$$EmpID \rightarrow EmpName, DeptID, ManagerID$$

$$DeptID \rightarrow DeptName$$

$$ManagerID \rightarrow ManagerName$$

Therefore, the transitive dependencies are:

$$EmpID \rightarrow DeptID \rightarrow DeptName$$

and

$$EmpID \rightarrow ManagerID \rightarrow ManagerName$$

Candidate Key

EmpID

3NF DECOMPOSITION
EMPLOYEE

EMPLOYEE(EmpID, EmpName, DeptID, ManagerID)

DEPARTMENT

DEPARTMENT(DeptID, DeptName)

MANAGER

MANAGER(ManagerID, ManagerName)

This removes the transitive dependencies.

CREATE TABLE AND INSERT DATA

```
mysql> CREATE TABLE Department (
    ->     DeptID INT PRIMARY KEY,
    ->     DeptName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Department
    -> VALUES
    -> (10, 'Finance'),
    -> (20, 'IT'),
    -> (30, 'HR');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Manager (
    ->     ManagerID INT PRIMARY KEY,
    ->     ManagerName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Manager
    -> VALUES
    -> (201, 'John'),
    -> (202, 'David'),
    -> (203, 'Sarah');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Employee (
    ->     EmpID INT PRIMARY KEY,
    ->     EmpName VARCHAR(50),
    ->     DeptID INT,
    ->     ManagerID INT,
    ->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID),
    ->     FOREIGN KEY (ManagerID) REFERENCES Manager(ManagerID)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO Employee
    -> VALUES
    -> (101, 'Rahul', 10, 201),
    -> (102, 'Priya', 20, 202),
    -> (103, 'Arun', 20, 202),
    -> (104, 'Sneha', 30, 203);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Employee;
+-------+---------+--------+-----------+
| EmpID | EmpName | DeptID | ManagerID |
+-------+---------+--------+-----------+
|   101 | Rahul   |     10 |       201 |
|   102 | Priya   |     20 |       202 |
|   103 | Arun    |     20 |       202 |
|   104 | Sneha   |     30 |       203 |
+-------+---------+--------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Employee;
+-------+---------+--------+-----------+
| EmpID | EmpName | DeptID | ManagerID |
+-------+---------+--------+-----------+
|   101 | Rahul   |     10 |       201 |
|   102 | Priya   |     20 |       202 |
|   103 | Arun    |     20 |       202 |
|   104 | Sneha   |     30 |       203 |
+-------+---------+--------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Department;
+--------+----------+
| DeptID | DeptName |
+--------+----------+
|     10 | Finance  |
|     20 | IT       |
|     30 | HR       |
+--------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Manager;
+-----------+-------------+
| ManagerID | ManagerName |
+-----------+-------------+
|       201 | John        |
|       202 | David       |
|       203 | Sarah       |
+-----------+-------------+
```

```
mysql> SELECT
    ->     E.EmpID,
    ->     E.EmpName,
    ->     D.DeptName,
    ->     M.ManagerName
    -> FROM Employee E
    -> JOIN Department D
    ->     ON E.DeptID = D.DeptID
    -> JOIN Manager M
    ->     ON E.ManagerID = M.ManagerID;
+-------+---------+----------+-------------+
| EmpID | EmpName | DeptName | ManagerName |
+-------+---------+----------+-------------+
|   101 | Rahul   | Finance  | John        |
|   102 | Priya   | IT       | David       |
|   103 | Arun    | IT       | David       |
|   104 | Sneha   | HR       | Sarah       |
+-------+---------+----------+-------------+
4 rows in set (0.00 sec)
```

Justification
Each non-key attribute in the decomposed relations depends directly on its respective key. This prevents repeated department and manager information and avoids insertion, deletion, and update anomalies.

6) Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.

Given Relation

LIBRARY(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)

Assume a member can borrow multiple books and the library can have multiple independent authors/genres associated with books.

The important dependency is:

MemberID → MemberName

For the 4NF case, assume each member can have multiple books, independently of another multi-valued fact such as multiple genres of interest.

A cleaner practical relation for demonstrating the MVD is:

MEMBER_LIBRARY(MemberID, BookID, Genre)
with:

MemberID →→ BookID
MemberID →→ Genre

These are multi-valued dependencies.

Example of Redundancy

Suppose Member 101 has:

- **Books: B01, B02**
- **Genres: Fiction, Science**

The relation may contain:

MemberID	BookID	Genre
101	B01	Fiction
101	B01	Science
101	B02	Fiction
101	B02	Science

The combinations create unnecessary redundancy**.**

4NF DECOMPOSITION

Separate the independent multi-valued facts.

MEMBER_BOOK

MEMBER_BOOK(MemberID, BookID)

MEMBER_GENRE

MEMBER_GENRE(MemberID, Genre)

CREATE TABLE AND INSERT DATA

```
mysql> CREATE TABLE Member_Library (
    ->     MemberID INT,
    ->     BookID VARCHAR(10),
    ->     Genre VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Member_Library
    -> VALUES
    -> (101, 'B01', 'Fiction'),
    -> (101, 'B01', 'Science'),
    -> (101, 'B02', 'Fiction'),
    -> (101, 'B02', 'Science'),
    -> (102, 'B03', 'History'),
    -> (102, 'B04', 'History');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Member_Library;
+----------+--------+---------+
| MemberID | BookID | Genre   |
+----------+--------+---------+
|      101 | B01    | Fiction |
|      101 | B01    | Science |
|      101 | B02    | Fiction |
|      101 | B02    | Science |
|      102 | B03    | History |
|      102 | B04    | History |
|      101 | B01    | Fiction |
|      101 | B01    | Science |
|      101 | B02    | Fiction |
|      101 | B02    | Science |
|      102 | B03    | History |
|      102 | B04    | History |
+----------+--------+---------+
12 rows in set (0.00 sec)

mysql> CREATE TABLE Member_Book (
    ->     MemberID INT,
    ->     BookID VARCHAR(10),
    ->     PRIMARY KEY (MemberID, BookID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Member_Book
    -> VALUES
    -> (101, 'B01'),
    -> (101, 'B02'),
    -> (102, 'B03'),
    -> (102, 'B04');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Member_Genre (
    ->     MemberID INT,
    ->     Genre VARCHAR(30),
    ->     PRIMARY KEY (MemberID, Genre)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Member_Genre
    -> VALUES
    -> (101, 'Fiction'),
    -> (101, 'Science'),
    -> (102, 'History');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Member_Book;
+----------+--------+
| MemberID | BookID |
+----------+--------+
|      101 | B01    |
|      101 | B02    |
|      102 | B03    |
|      102 | B04    |
+----------+--------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Member_Genre;
+----------+---------+
| MemberID | Genre   |
+----------+---------+
|      101 | Fiction |
|      101 | Science |
|      102 | History |
+----------+---------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     MB.MemberID,
    ->     MB.BookID,
    ->     MG.Genre
    -> FROM Member_Book MB
    -> JOIN Member_Genre MG
    -> ON MB.MemberID = MG.MemberID;
+----------+--------+---------+
| MemberID | BookID | Genre   |
+----------+--------+---------+
|      101 | B01    | Fiction |
|      101 | B02    | Fiction |
|      101 | B01    | Science |
|      101 | B02    | Science |
|      102 | B03    | History |
|      102 | B04    | History |
+----------+--------+---------+
6 rows in set (0.00 sec)
```

Justification

4NF requires every non-trivial multi-valued dependency to have a superkey as its determinant. By separating the independent multi-valued attributes into two relations, the MVDs are removed and unnecessary combinations are avoided.

7) Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.

Consider the telecom billing relation:

BILLING(BillID, CustomerID, CustomerName, PlanID, PlanName, Amount, BillDate)

Functional Dependencies

BillID → CustomerID, PlanID, Amount, BillDate

CustomerID → CustomerName

PlanID → PlanName

Therefore:

BillID → CustomerID → CustomerName

And

BillID → PlanID → PlanName

are transitive dependencies.

Candidate Key
BillID uniquely identifies each bill.

Candidate Key=BillID

Primary Key

BillID

CREATE TABLE AND INSERT DATA

```
Database changed
mysql> CREATE TABLE Billing (
    ->     BillID INT PRIMARY KEY,
    ->     CustomerID INT,
    ->     CustomerName VARCHAR(50),
    ->     PlanID VARCHAR(10),
    ->     PlanName VARCHAR(50),
    ->     Amount DECIMAL(10,2),
    ->     BillDate DATE
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Billing
    -> VALUES
    -> (1001, 101, 'Rahul', 'P01', '5G Plan', 599, '2026-08-01'),
    -> (1002, 102, 'Priya', 'P02', 'Family Plan', 799, '2026-08-02'),
    -> (1003, 101, 'Rahul', 'P01', '5G Plan', 599, '2026-08-03'),
    -> (1004, 103, 'Arun', 'P03', 'Basic Plan', 399, '2026-08-04');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Billing;
+--------+------------+--------------+--------+-------------+--------+------------+
| BillID | CustomerID | CustomerName | PlanID | PlanName    | Amount | BillDate   |
+--------+------------+--------------+--------+-------------+--------+------------+
|   1001 |        101 | Rahul        | P01    | 5G Plan     | 599.00 | 2026-08-01 |
|   1002 |        102 | Priya        | P02    | Family Plan | 799.00 | 2026-08-02 |
|   1003 |        101 | Rahul        | P01    | 5G Plan     | 599.00 | 2026-08-03 |
|   1004 |        103 | Arun         | P03    | Basic Plan  | 399.00 | 2026-08-04 |
+--------+------------+--------------+--------+-------------+--------+------------+
4 rows in set (0.00 sec)
```

Justification

BillID uniquely identifies every billing record, so it is the candidate key and can be selected as the primary key. CustomerID determines customer information and PlanID determines plan information, creating transitive dependencies from BillID. These dependencies should be considered during normalization to remove redundancy.

8) A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.

Given Relation

PRODUCT(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)

Assume:

PID → PName, SupplierID, Category, Price, Warehouse

and:

SupplierID → SupplierName

Therefore:

PID → SupplierID → SupplierName

is a transitive dependency.

Candidate Key

PID

NORMALIZATION

Separate supplier information:

PRODUCT

PRODUCT(PID, PName, SupplierID, Category, Price, Warehouse)

SUPPLIER

SUPPLIER(SupplierID, SupplierName)

Now SupplierID is the primary key of SUPPLIER and a foreign key in PRODUCT.

This gives a normalized 3NF/BCNF design under the stated dependencies.

CREATE AND INSERT DATA

```
mysql> CREATE TABLE Supplier (
    ->     SupplierID INT PRIMARY KEY,
    ->     SupplierName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Supplier
    -> VALUES
    -> (201, 'ABC Suppliers'),
    -> (202, 'Global Traders'),
    -> (203, 'Fresh Supplies');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Product (
    ->     PID INT PRIMARY KEY,
    ->     PName VARCHAR(50),
    ->     SupplierID INT,
    ->     Category VARCHAR(50),
    ->     Price DECIMAL(10,2),
    ->     Warehouse VARCHAR(50),
    ->     FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Product
    -> VALUES
    -> (101, 'Laptop', 201, 'Electronics', 65000, 'Chennai'),
    -> (102, 'Mouse', 202, 'Accessories', 800, 'Bangalore'),
    -> (103, 'Keyboard', 201, 'Accessories', 1500, 'Chennai'),
    -> (104, 'Monitor', 203, 'Electronics', 12000, 'Hyderabad');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Supplier;
+------------+----------------+
| SupplierID | SupplierName   |
+------------+----------------+
|        201 | ABC Suppliers  |
|        202 | Global Traders |
|        203 | Fresh Supplies |
+------------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Product;
+-----+----------+------------+-------------+----------+------------+
| PID | PName    | SupplierID | Category    | Price    | Warehouse  |
+-----+----------+------------+-------------+----------+------------+
| 101 | Laptop   |        201 | Electronics | 65000.00 | Chennai    |
| 102 | Mouse    |        202 | Accessories |   800.00 | Bangalore  |
| 103 | Keyboard |        201 | Accessories |  1500.00 | Chennai    |
| 104 | Monitor  |        203 | Electronics | 12000.00 | Hyderabad  |
+-----+----------+------------+-------------+----------+------------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     P.PID,
    ->     P.PName,
    ->     P.SupplierID,
    ->     S.SupplierName,
    ->     P.Category,
    ->     P.Price,
    ->     P.Warehouse
    -> FROM Product P
    -> JOIN Supplier S
    -> ON P.SupplierID = S.SupplierID;
+-----+----------+------------+----------------+-------------+----------+------------+
| PID | PName    | SupplierID | SupplierName   | Category    | Price    | Warehouse  |
+-----+----------+------------+----------------+-------------+----------+------------+
| 101 | Laptop   |        201 | ABC Suppliers  | Electronics | 65000.00 | Chennai    |
| 103 | Keyboard |        201 | ABC Suppliers  | Accessories |  1500.00 | Chennai    |
| 102 | Mouse    |        202 | Global Traders | Accessories |   800.00 | Bangalore  |
| 104 | Monitor  |        203 | Fresh Supplies | Electronics | 12000.00 | Hyderabad  |
+-----+----------+------------+----------------+-------------+----------+------------+
4 rows in set (0.00 sec)
```

Justification

The original relation stores supplier information repeatedly for every product supplied by the same supplier. Since SupplierID → SupplierName, SupplierName has a transitive dependency through SupplierID. Separating the Supplier relation removes this redundancy and gives a normalized 3NF/BCNF design under the stated functional dependencies.

9) Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.

Consider:

SCHOOL(
StudentID, StudentName,
ClassID, ClassName,
TeacherID, TeacherName,
SubjectID, SubjectName
)

Functional Dependencies

StudentID → StudentName, ClassID
ClassID → ClassName, TeacherID
TeacherID → TeacherName
SubjectID → SubjectName

Thus, transitive dependencies include:

StudentID → ClassID → ClassName, TeacherID → TeacherName

Candidate Key

Assume a student can study multiple subjects:

(StudentID,SubjectID)

BCNF DECOMPOSITION
STUDENT

STUDENT(StudentID, StudentName, ClassID)

CLASS

CLASS(ClassID, ClassName, TeacherID)

TEACHER

TEACHER(TeacherID, TeacherName)

SUBJECT

SUBJECT(SubjectID, SubjectName)

ENROLLMENT

ENROLLMENT(StudentID, SubjectID)

Each determinant is a key in its respective relation, satisfying BCNF.

CREATE TABLE AND INSERT DATA

```
mysql> CREATE TABLE Teacher (
    ->     TeacherID INT PRIMARY KEY,
    ->     TeacherName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Teacher VALUES
    -> (201, 'Kumar'),
    -> (202, 'Priya'),
    -> (203, 'Ravi');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Class (
    ->     ClassID INT PRIMARY KEY,
    ->     ClassName VARCHAR(50),
    ->     TeacherID INT,
    ->     FOREIGN KEY (TeacherID) REFERENCES Teacher(TeacherID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Class VALUES
    -> (1, 'CSE-A', 201),
    -> (2, 'CSE-B', 202),
    -> (3, 'ECE-A', 203);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Student (
    ->     StudentID INT PRIMARY KEY,
    ->     StudentName VARCHAR(50),
    ->     ClassID INT,
    ->     FOREIGN KEY (ClassID) REFERENCES Class(ClassID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Student VALUES
    -> (101, 'Rahul', 1),
    -> (102, 'Priya', 2),
    -> (103, 'Arun', 1);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Subject (
    ->     SubjectID INT PRIMARY KEY,
    ->     SubjectName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Subject VALUES
    -> (301, 'DBMS'),
    -> (302, 'Python'),
    -> (303, 'Java');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Enrollment (
    ->     StudentID INT,
    ->     SubjectID INT,
    ->     PRIMARY KEY (StudentID, SubjectID),
    ->     FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    ->     FOREIGN KEY (SubjectID) REFERENCES Subject(SubjectID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Enrollment VALUES
    -> (101, 301),
    -> (101, 302),
    -> (102, 301),
    -> (103, 303);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT
    ->     S.StudentID,
    ->     S.StudentName,
    ->     C.ClassName,
    ->     T.TeacherName,
    ->     SU.SubjectName
    -> FROM Student S
    -> JOIN Class C ON S.ClassID = C.ClassID
    -> JOIN Teacher T ON C.TeacherID = T.TeacherID
    -> JOIN Enrollment E ON S.StudentID = E.StudentID
    -> JOIN Subject SU ON E.SubjectID = SU.SubjectID;
+-----------+-------------+-----------+-------------+-------------+
| StudentID | StudentName | ClassName | TeacherName | SubjectName |
+-----------+-------------+-----------+-------------+-------------+
|       101 | Rahul       | CSE-A     | Kumar       | DBMS        |
|       101 | Rahul       | CSE-A     | Kumar       | Python      |
|       102 | Priya       | CSE-B     | Priya       | DBMS        |
|       103 | Arun        | CSE-A     | Kumar       | Java        |
+-----------+-------------+-----------+-------------+-------------+
4 rows in set (0.01 sec)
```

Justification

The original School ERP relation contains repeated student, class, teacher, and subject information. Decomposing it into separate entities removes the transitive dependencies and stores each fact only once. The resulting relations have determinants that are keys, giving a BCNF design.

10) Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.

Consider:

STREAMING(MovieID, ActorID, PlatformID)

Assume a movie can independently have:

- multiple actors
- multiple streaming platforms

For a 5NF example, assume the valid relationship is represented through three independent pairwise relationships.

Join Dependency

∗(Movie,Actor),(Movie,Platform),(Actor,Platform)

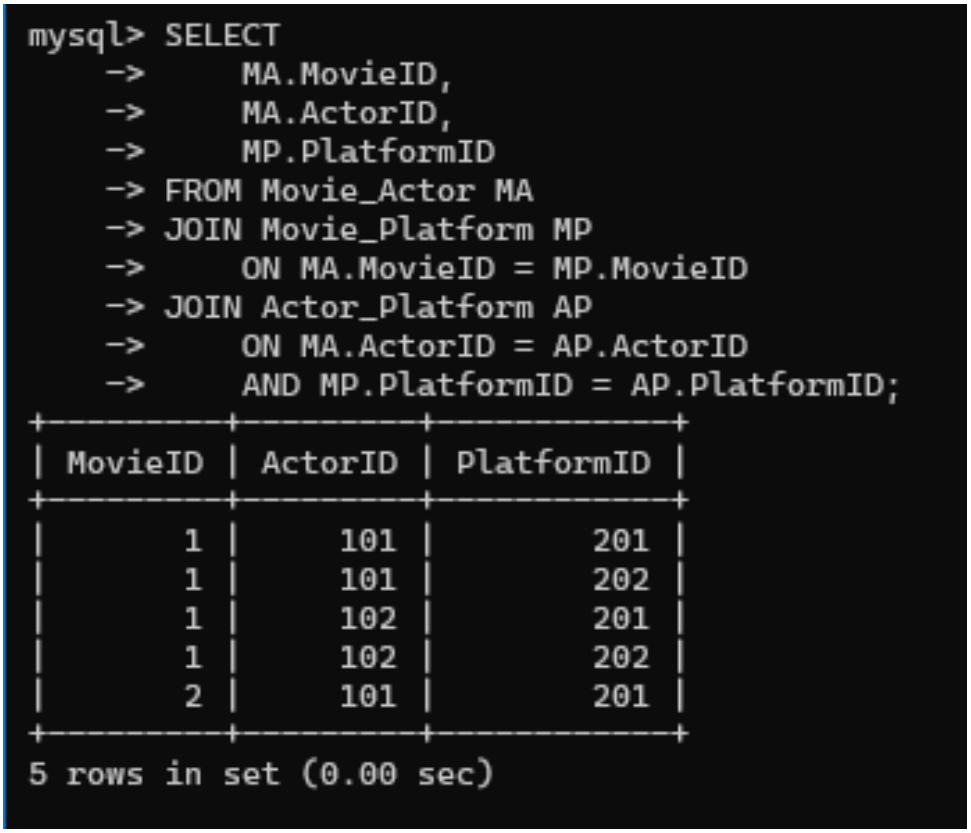
So decompose into:

MOVIE_ACTOR(MovieID, ActorID)
MOVIE_PLATFORM(MovieID, PlatformID)
ACTOR_PLATFORM(ActorID, PlatformID)

CREATE TABLES AND INSERT DATA

```
Database changed
mysql> CREATE TABLE Movie_Actor (
    ->     MovieID INT,
    ->     ActorID INT,
    ->     PRIMARY KEY (MovieID, ActorID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Movie_Actor VALUES
    -> (1, 101),
    -> (1, 102),
    -> (2, 101);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Movie_Platform (
    ->     MovieID INT,
    ->     PlatformID INT,
    ->     PRIMARY KEY (MovieID, PlatformID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Movie_Platform VALUES
    -> (1, 201),
    -> (1, 202),
    -> (2, 201);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Actor_Platform (
    ->     ActorID INT,
    ->     PlatformID INT,
    ->     PRIMARY KEY (ActorID, PlatformID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Actor_Platform VALUES
    -> (101, 201),
    -> (101, 202),
    -> (102, 201),
    -> (102, 202);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Movie_Actor;
+---------+---------+
| MovieID | ActorID |
+---------+---------+
|       1 |     101 |
|       1 |     102 |
|       2 |     101 |
+---------+---------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Movie_Platform;
+---------+------------+
| MovieID | PlatformID |
+---------+------------+
|       1 |        201 |
|       1 |        202 |
|       2 |        201 |
+---------+------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Actor_Platform;
+---------+------------+
| ActorID | PlatformID |
+---------+------------+
|     101 |        201 |
|     101 |        202 |
|     102 |        201 |
|     102 |        202 |
+---------+------------+
4 rows in set (0.00 sec)
```



```
mysql> SELECT
    ->     MA.MovieID,
    ->     MA.ActorID,
    ->     MP.PlatformID
    -> FROM Movie_Actor MA
    -> JOIN Movie_Platform MP
    ->     ON MA.MovieID = MP.MovieID
    -> JOIN Actor_Platform AP
    ->     ON MA.ActorID = AP.ActorID
    ->     AND MP.PlatformID = AP.PlatformID;
+---------+---------+------------+
| MovieID | ActorID | PlatformID |
+---------+---------+------------+
|       1 |     101 |        201 |
|       1 |     101 |        202 |
|       1 |     102 |        201 |
|       1 |     102 |        202 |
|       2 |     101 |        201 |
+---------+---------+------------+
5 rows in set (0.00 sec)
```

Justification

The original movie-streaming relation can contain redundancy when movie, actor, and platform relationships are stored together. Decomposing it into pairwise relations removes the join dependency redundancy. The successful JOIN reconstructs the required tuples, demonstrating a lossless-join decomposition into 5NF.

THANK YOU